

WOMEN SAFETY APP WITH SOS & LIVE TRACKING



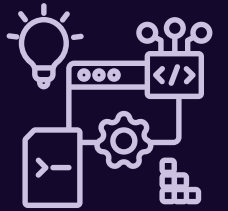
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Abstract

1. Provides a quick SOS mechanism with live GPS tracking and real-time alerts to trusted guardians during emergencies.
2. Uses AI-based risk prediction to identify potential safety risks and support proactive emergency response.
3. Includes a secure guardian dashboard for real-time monitoring, location tracking, and emergency management.

Tools & Technologies

- Kotlin + Android Studio – Mobile App
- Python + Machine Learning – AI Risk Prediction
- Firebase – Database & Real-time Data
- Fused Location API – Live GPS Tracking
- Twilio API – Emergency SMS & Calls
- React + JavaScript – Guardian Dashboard



Problem Statement & Objectives

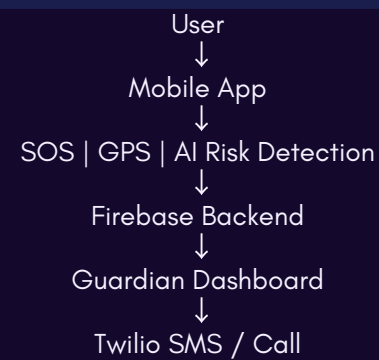
Problem:

- Emergency communication may be difficult or delayed.
- Lack of real-time location and risk monitoring.

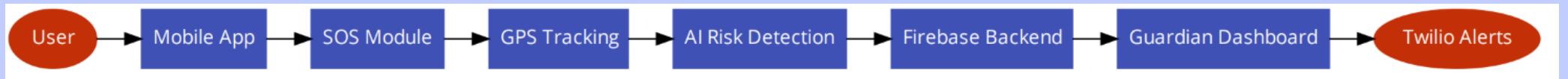
Objectives:

- Quick SOS + live GPS tracking.
- AI-based risk prediction.
- Real-time guardian alerts.

System Architecture



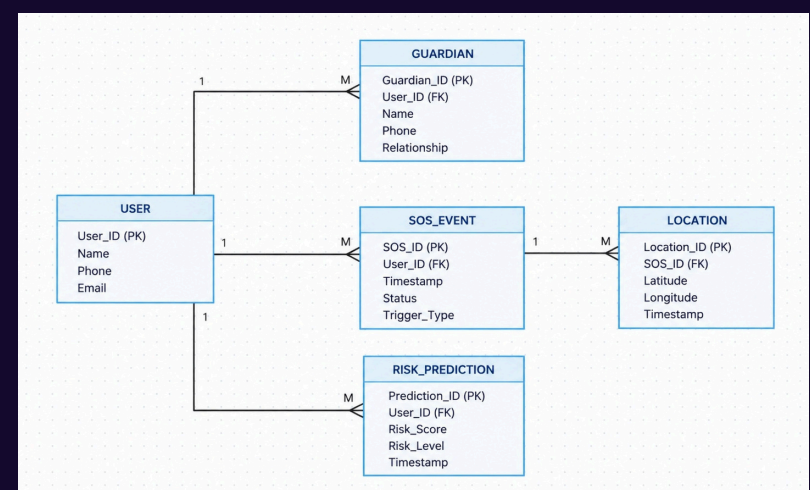
Architecture :



Existing vs Proposed System

Existing System	Proposed System
Manual emergency communication	Quick SOS alerts
Limited location tracking	Real-time GPS tracking
No AI risk detection	AI-based risk detection
No centralized monitoring	Guardian dashboard
Basic emergency response	SMS & call alerts via Twilio

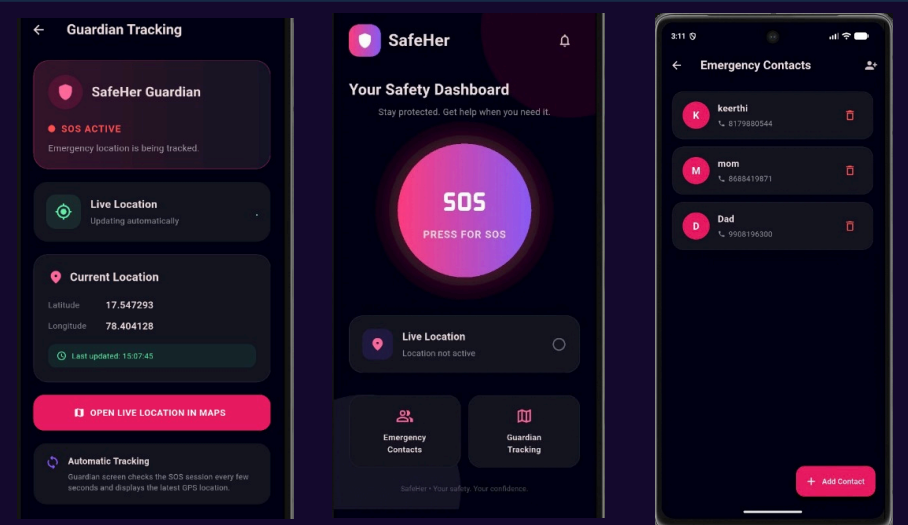
ER Diagram & Database Design



Feasibility Study

1. Technical: Uses available technologies like Kotlin, Firebase, GPS, AI/ML, and Twilio.
2. Economic: Uses mostly open-source and cloud-based tools with low development cost.
3. Operational: Simple mobile app and guardian dashboard for easy use.
4. Schedule: Can be developed and tested within the project timeline.
5. Security: Firebase authentication and secure data storage help protect user and emergency information.

Implementation Screenshots



Results & Testing

1. SOS alert triggered successfully.
2. Live GPS location tracked correctly.
3. AI risk level generated successfully.
4. Emergency SMS/call alerts tested.
5. Guardian dashboard displayed real-time updates.
6. Firebase data storage verified.

Conclusion

1. Provides a reliable SOS and live GPS safety solution.
2. Enables AI-based risk detection and real-time alerts.
3. Helps guardians monitor emergencies through a centralized dashboard.
4. Combines mobile, AI, GPS, and secure cloud technologies for women's safety.

Future Scope

1. • Smartwatch / wearable integration
2. • Voice-based SOS
3. • Offline support
4. • AI anomaly detection
5. • Emergency-service integration